The basal ganglia

The **basal ganglia** are a complicated group of subcortical brain areas thought to act as a gate to decision making and as a centre for processing rewards and their association with actions. One important aspect of the basal ganglia is how they relate to dopamine, one of the neuromodulators; the basal ganglia include the **substantia nigra**, one of the two main areas of the brain where **dopaminergic**, that is dopamine producing, neurons are found. The other area, the **Ventral Tegmental Area (VTA)** produces dopamine in reaction to rewarding events, or perhaps unexpectedly rewarding events; the prompt for release of dopamine by the substantia nigra is harder to summarise, but again, seems to be related to reward and reward-cued behaviours.

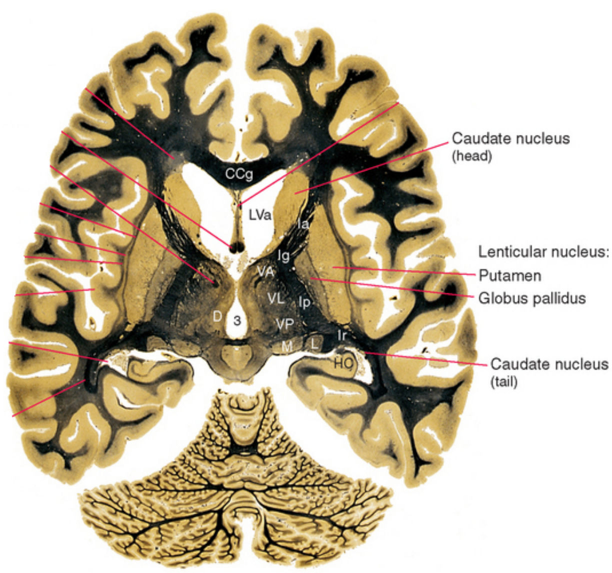

Figure 1: **Stained horizontal section through the human brain from *Nolte & Angevine: The Human Brain in Photographs and Diagrams (2007)*, with a subset of basal ganglia labelled.** This slice has been prepared with a stain that colours white matter black, leaving gray matter (concentrations of cell bodies) lighter in colour. The caudate, putamen, and globus pallidus are labelled. These nuclei are involved in selecting and gating actions, and in reward-based learning.

Parkinson's disease, a neurodegenerative disorder associated with stiff, even frozen, movements is associates with the loss of cells in the basal ganglia. It is believed that the basal ganglia provides a final 'go' signal allowing motor commands to travel to their muscle target and that with cell loss in the basal ganglia this 'go' signal does not occur. Providing **L-dopa**, a dopamine precursor, can alleviate this difficulty, though, of course, this broad uplift in dopamine is not a complete substitute for the more exquisitely modulated provision of dopamine we can assume is provided by the substantia nigra

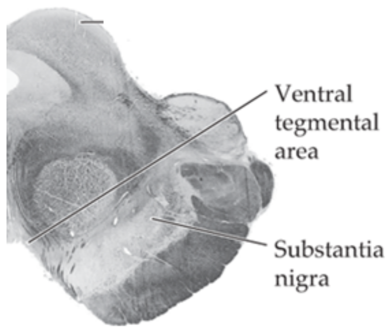

Figure 2: **Ventral tegmental area and substantia nigra**; Figure from J. Martin's *Neuroanatomy Text and Atlas*: This figure shows a secton of the midbrain at the level of the VTA and substantia nigra. In an un-stained preparation, the *substantia nigra* ("black stuff") is usually darker, but the slide has been prepared with a stain that colours white matter dark, leaving grey matter (nuclei with neuronal cell bodies) light in color.

Summary

The basal ganglia play a role in decision making and Parkinson's disease is associated with loss of neurons in the basal ganglia.

Just for fun

Some of the Latin names for brain areas are mildly amusing (apart from the substantia nigra, this content is not on the exam):

Substantia nigra: black stuff.

Globus pallidus: pale ball.

Substantia gelatinosa: jelly-like stuff (part of the spinal cord that may be involved in sensation/pain gating and integration).

Substantia innominata: untitled stuff (contains multiple systems, e.g. *nucleus basalis magnocellularis*, with contains neurons that release acetylcholine).

Locus coeruleus: blue place (source of axons that release the neuromodulator noradrenaline).